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Journal of Psychiatric Research 165 (2023) 298–304

Contents lists available at ScienceDirect

Journal of Psychiatric Research

journal homepage: www.elsevier.com/locate/jpsychires

Altered levels of fecal short-chain fatty acids are associated with subclinical inflammation and worse cognitive performance in patients with schizophrenia

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ARTICLE INFO

Keywords:
Psychosis
Gut-brain axis
Immunity
Cognition

ABSTRACT

Schizophrenia is a multi-systemic disorder that is associated with lipid profile disturbances, altered glucose homeostasis and subclinical inflammation. It has been proposed that dysfunction of the gut-brain axis might underlie these alterations. Short-chain fatty acids (SCFAs) are considered to play a pivotal role in the gut-brain axis. In this study, we aimed to compare fecal levels of SCFAs in patients with schizophrenia and healthy controls (HCs), taking into consideration their relationship with common peripheral blood alterations observed in schizophrenia. The study included 100 stable outpatients with schizophrenia and 55 HCs. The levels of SCFAs (acetic acid, propionic acid, butyric acid, isobutyric acid, valeric acid, isovaleric acid, and lactic acid) in fecal samples were measured. Also, lipid profile together with the levels of C-reactive protein, glucose and insulin were determined. The levels of isovaleric acid were significantly higher in patients with schizophrenia after co-varying for age, sex, and the adherence to the Mediterranean diet. Moreover, there were significant positive correlations of the levels of valeric acid, isovaleric acid and CRP in patients with schizophrenia. In this group of participants, higher levels of isovaleric acid were associated with significantly lower scores of delayed memory after adjustment for potential covariates and interactions with CRP levels. Our results indicate that individuals with schizophrenia show altered levels of isovaleric acid that might be associated with impairments of delayed memory. The association with cognitive impairments might be independent of interactions with immune-inflammatory processes. Longitudinal and experimental studies are needed to test causal mechanisms of observed correlations.

1. Introduction

Schizophrenia is a chronic and severe mental illness that affects millions of people worldwide (Tandon et al., 2008). It is characterized by a range of symptoms including hallucinations, delusions, and cognitive impairment, which can significantly impact an individual ability to function in daily life (Tripathi et al., 2018). Cognitive impairment is a common and debilitating symptom of schizophrenia, and it is associated with a range of negative outcomes including poor functional outcome and reduced quality of life (Green en Harvey, 2014). While the exact causes of schizophrenia are not fully understood, research suggests that both genetic and environmental factors may play a role in its development (Kendler et al., 2003).

There is a growing interest in the role of the gut-brain axis in the pathophysiology of schizophrenia. One potential factor that has been implicated in schizophrenia is inflammation, which is a normal response of the body to injury or infection (Miller en Wolin, 1996). However, chronic inflammation has been linked to a range of negative health outcomes, including cognitive decline and an increased risk of developing mental disorders such as schizophrenia (Kirkpatrick en Miller and

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https://doi.org/10.1016/j.jpsychires.2023.07.042
Received 21 January 2023; Received in revised form 15 June 2023; Accepted 30 July 2023
Available online 31 July 2023
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Wolin, 2013). Emerging evidence suggests that the gut microbiome, which is the collection of microorganisms that inhabit the digestive tract, may play a role in brain function and mental health (Cryan en Dinan, 2012). Dysbiosis, or an imbalance in the gut microbiome, has been linked to a variety of negative health outcomes, including inflammation and cognitive decline (Ettinger, 2022). However, the relationship between the gut microbiome and cognitive function in schizophrenia has not been sufficiently explored.

Bioactive short-chain fatty acids (SCFAs) are saturated fatty acids with a carbon chain length ranging from one to six carbon atoms. Among SCFAs, acetate, propionate, butyrate are predominant in the intestinal tract (Cummings et al., 1987). They are products of bacterial fermentation of dietary fiber in the gastrointestinal tract (Miller en Wolin, 1996). Additionally, specific bacteria representing gut microbiota produce lactate, which is the most common short-chain hydroxy–fatty acid in the intestinal lumen. It can be converted to other SCFAs by a subgroup of lactate-fermenting bacterial species (Louis et al., 2022). Previous studies have suggested that SCFAs may exert anti-inflammatory effects (Han et al., 2018) Moreover, SCFAs have also been found to regulate maturation of microglia that serve as important component of the immune-inflammatory response in the central nervous system (Erny et al., 2015). It has been also revealed that administration of SCFAs in rodents may alter the levels of gastrointestinal hormones that regulate food intake, energy expenditure and glucose metabolism. These include pancreatic polypeptide, glucagon-like peptide-1, leptin, ghrelin, and insulin (Dalile et al., 2019). These observations suggest that SCFAs might be related to metabolic syndrome components that are highly prevalent in patients with schizophrenia (Mitchell et al., 2013).

In parallel, recent systematic reviews show that increased levels of lactic-acid bacteria and low levels of bacteria producing SCFAs are observed in patients with severe mental disorders (Borkent et al., 2022; McGuinness et al., 2022). Nonetheless, the involvement of SCFAs in the pathophysiology of schizophrenia remains unclear. Indeed, only one study explored the levels of SCFAs in the peripheral blood in drug-naïve individuals with first-episode psychosis showing that cognitive dysfunction and abnormal serum levels of SCFAs occur in the early phase of schizophrenia (Li et al., 2022). Another one focused on the levels of SCFAs in fecal samples (Liang et al., 2022). The authors demonstrated that in patients with schizophrenia, microbial metabolism is dominated by protein fermentation. However, both studies either did not investigate the associations of SCFAs levels with cognition or the peripheral blood markers. Taking into consideration these research gaps, we aimed to compare fecal levels of SCFAs between patients with schizophrenia and healthy controls. Moreover, we explored the relationship of fecal levels of SCFAs with the peripheral blood indices of inflammation, lipid and glucose metabolism as well as cognitive performance in both groups of participants.

2. Material and methods

2.1. Participants

The present study included 100 outpatients with schizophrenia and 55 healthy controls. Participants were enrolled at two university hospitals (Wroclaw and Szczecin, Poland). Patients were recruited if they met the following criteria: 1) aged 18–65 years; 2) a diagnosis of schizophrenia according to the DSM-IV criteria, confirmed with the use of Operational Criteria for Psychotic Illness (OPCRIT) checklist (McGuffin, 1991); 3) remission of positive and disorganization symptoms according to the criteria referring to assessment with the Positive and Negative Syndrome Scale (PANSS) (Kay et al., 1987) and proposed by Andreasen (1995); 4) stable psychopharmacological treatment over the period of preceding 6 months; 5) comorbid substance dependence (except of nicotine and caffeine); 6) intellectual disability based on medical records and 7) clinical symptoms of infectious disease. The dosage of antipsychotics was converted to chlorpromazine equivalents (CPZeq). The use of other psychiatric medications was also recorded.

Healthy controls were recruited from the local community. Negative current and lifetime history of mental disorders was confirmed using the Mini-International Neuropsychiatic Interview (M.I.N.I.) (Sheehan et al., 1998). No first- and second-degree relatives affected by mood and psychotic disorders were reported by healthy controls.

The study received approval of Bioethics Committees at Wroclaw Medical University (Wroclaw, Poland, approval number: 512/2019) and Pomeranian Medical University (Szczecin, Poland, approval number: KB-0012/130/2019). All patients and healthy controls signed written informed consent.

2.2. Assessment of clinical manifestation

On the day of recruitment, all patients received assessment with the following scales: 1) the Positive and Negative Syndrome Scale (PANSS); 2) the Calgary Depression Scale for Schizophrenia (CDSS) (Addington et al., 1993) and 3) the Social and Occupational Functioning Assessment Scale (SOFAS) (Hall, 1995). All participants underwent examination of cognitive performance using the Repeatable Battery for the Assessment of Neuropsychological Status (RBANS). The RBANS is a set of tasks measuring performance of immediate memory, visuospatial/constructional abilities, language, attention and delayed memory. Higher scores indicate better cognitive performance (Randolph et al., 1998).

2.3. Dietary habits

The Polish version of the Food Frequency Questionnaire (FFQ-6) was administered to record dietary intake of various food products over the preceding 12 months. The FFQ-6 is a 62-item interview, and each item measures the frequency of food consumption according to a 6-point scale (scored from 1 – “never or almost never” to 6 – “several times a day”) (Niedzwiedzka et al., 2019). In the present study, data collected using the FFQ-6 were the basis to assess adherence to the Mediterranean diet using the aMED score (Krusinska et al., 2018). The following food categories were used to calculate the aMED score: 1) vegetables; 2) fruits; 3) whole grains; 4) fish; (5) legumes; (6) nuts and seeds; 7) the ratio of vegetable oils to animal fat and 8) red and processed meat.

2.4. Analysis of fecal samples

The analysis of SCFAs in stool extracts was performed as previously described (Torii et al., 2010). Briefly, stool samples were extracted with 70%-ethanol. Debris was removed by centrifugation, and 500 μL of supernatant was transferred to a new tube that was mixed with 50 μL of internal standard (2-ethylbutyric acid, 200 mM in 50%-aqueous methanol), 300 μL of dehydrated pyridine (Merck) 3% vol/vol (v/v) in ethanol, 300 μL of 250 mM N-(3-dimethlaminopropyl)-N′-ethylcarbodiimide hydrochloride (Sigma-Aldrich) in ethanol, and 300 μL of 20 mM 2-nitrophenylhydrazine hydrochloride (Sigma-Aldrich) in ethanol. Samples were incubated at 60 °C for 20 min And mixed with 200 μl of sodium hydroxide solutions (15% weight/volume with water, and used at a sodium hydroxide solution/methanol ratio of 80/20, v/v) as a reaction termination. After cooling, the mixture was extracted with 2 ml of phosphoric acid aqueous solution (0.5 mol/L) and 4 ml of diethyl ether twice. The organic phase was collected and extracted with water. The organic phase containing fatty acid hydrazides was collected, evaporated and dissolved in 150 μl of methanol, and subjected to the high-performance liquid chromatography (HPLC) analysis.

The HPLC was performed using the 1525 Binary HPLC Pump with the 2489 UV/Visible (UV/Vis) Detector (Waters) and the C18 column (Shimadzu 5 μm C18-120 Å, 250 × 4.6 mm), with mobile phase acetonitrile-methanol-water (30:16:54). The column temperature was 50 °C; the flow rate was set at 1 ml/min and the wavelength was 400 nm (Arpaia et al., 2013; Torii et al., 2010).

2.5. Biochemical parameters

All biochemical parameters were analyzed in fasting serum samples. The levels of high-sensitivity C-reactive protein (CRP) were determined using the immunoturbidometric assay (Cobas8000c/701, Roche). Colorimetric methods were implemented to analyze the levels of total cholesterol, lipoproteins and triglycerides. The electrochemiluminescence assay and spectrophotometric methods were used to determine the concentrations of insulin and glucose, respectively. The Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) was calculated using the formula: insulin × glucose (mg/dl)/405 (Matthews et al., 1985).

2.6. Data analysis

Differential distribution of categorical data was tested using the χ^2 test. Differences in continuous variables between both groups of participants were analyzed by the *t*-test (in case of normal distribution) and the Mann-Whitney *U* test (in case of non-normal distribution). The analysis of co-variance (ANCOVA) was carried out to further assess significant differences in the levels of SCFAs after controlling for the effects of age, sex, BMI, and the aMED score. Correlations of the levels of lactic acid and SCFAs with other continuous variables were assessed using the Spearman rank correlation coefficients. In case of significant bivariate correlations of the levels of SCFAs and lactic acid with cognitive performance, multiple linear regression analyses were performed to control for the effects of potential confounding factors. To perform this part of data analysis, the measures of cognitive performance were transformed to z-scores and included as dependent variables in separate models. Independent variables were tested within separate blocks that covered main and interaction effects with the group status (individuals with schizophrenia vs. healthy controls). Regression lines were plotted for the model with the largest change of the R^2 values and a significant change of the F value. Covariates were selected from variables showing significant between-group differences and correlations with performance on a specific cognitive domain. To avoid potential multicollinearity, independent continuous variables were also converted to z-scores. Multicollinearity was analyzed by calculating the variance inflation factor (VIF). The VIF values lower than 10 were interpreted as indicating no significant multicollinearity (Vittinghoff et al., 2005; James et al., 2017). To control for multiple bivariate comparisons, the Benjamini-Hochberg correction was applied to p–values from bivariate tests. In case of the ANCOVA and multiple linear regression analyses, the level of significance was set at $p < 0.05$. Analyses were carried out in the SPSS software, version 28.

3. Results

Both groups of participants did not differ significantly in terms of age, sex as well as the level of own and parental education (Table 1). Patients with schizophrenia had significantly higher body mass index (BMI). They showed significantly higher level of physical addiction to nicotine and lower aMED score. As expected, individuals with schizophrenia performed significantly lower on all RBANS domains.

The levels of triglycerides, CRP, HOMA-IR and isovaleric acid were significantly higher in the group of patients compared to healthy controls (Table 2). In turn, the levels of HDL were significantly lower in patients with schizophrenia. The ANCOVA revealed that the level of isovaleric acid was significantly higher (Fig. 1A) in patients with schizophrenia ($F = 3.906$, $p = 0.048$) after co-varying for age ($F = 6.532$, $p = 0.012$), sex ($F = 0.237$, $p = 0.627$), BMI ($F = 0.025$, $p = 0.876$) and the aMED score ($F = 0.610$, $p = 0.436$).

Bivariate correlations of the levels of lactic acid and SCFAs with other metabolic parameters and dietary habits are reported in Table 3. In patients with schizophrenia, higher levels of valeric acid and isovaleric acid were significantly associated with higher levels of CRP. Also, a significant negative correlation between the levels of acetic acid and the levels of LDL was found in patients with schizophrenia. In turn, among healthy controls, higher levels of CRP were significantly associated with higher levels of propionic acid, acetic acid, butyric and isobutyric acid. Moreover, there were significant positive correlations of the levels of acetic acid, valeric acid, butyric and isobutyric acid with the levels of triglycerides in healthy controls.

Correlations of the levels of lactic acid and SCFAs with clinical characteristics are presented in Table 4. Higher levels of isovaleric acid

Table 1
General and clinical characteristics of participants.

	Schizophrenia (n = 100)	Healthy controls (n = 55)	p
Age, years	43.7 ± 12.0	42.3 ± 15.0	0.410
Sex, males	60 (60.0)	30 (54.5)	0.510
Education, higher	22 (22.0)	9 (16.4)	0.401
Paternal education, higher	24 (24.0)	12 (21.8)	0.758
Maternal education, higher	18 (18.0)	8 (14.5)	0.582
BMI, kg/m^2	28.6 ± 5.3	26.4 ± 4.9	0.012*
FTND	2.4 ± 3.2	0.5 ± 1.5	<0.001*
aMED	2.3 ± 1.3	3.4 ± 1.4	<0.001*
RBANS, immediate memory	36.4 ± 10.1	47.0 ± 6.4	<0.001*
RBANS, visuospatial/ constructional	33.9 ± 5.6	37.4 ± 3.2	<0.001*
RBANS, language	25.1 ± 5.2	30.4 ± 5.0	<0.001*
RBANS, attention	43.2 ± 14.6	58.8 ± 10.2	<0.001*
RBANS, delayed memory	40.5 ± 10.8	50.9 ± 7.2	<0.001*
Illness duration, years	18.8 ± 12.3	–	–
PANSS, negative symptoms	16.0 ± 9.1	–	–
CDSS, depressive symptoms	1.6 ± 3.0	–	–
SOFAS, general functioning	59.1 ± 20.7	–	–
CPZeq, mg/day	608.0 ± 373.5	–	–
Antidepressants, yes	9 (90.0)	–	–
Mood stabilizers, yes	10 (10.0)	–	–

*Significant differences after the Benjamini-Hochberg correction.
The aMED score was not available in case of 4 individuals with schizophrenia.
Abbreviations: aMED, adherence to Mediterranean diet; BMI, body mass index; CDSS, the Calgary Depression Scale for Schizophrenia; CPZeq; chlorpromazine equivalent dosage; CRP, C-reactive protein; FTND, the Fagerstrom Test of Nicotine Dependence; RBANS, the Repeatable Battery for the Assessment of Neuropsychological Status.

Table 2
The levels of biochemical parameters.

	Schizophrenia (n = 100)	Healthy controls (n = 55)	p
LDL, mg/dl	110.9 ± 33.9	109.0 ± 27.3	0.894
HDL, mg/dl	51.9 ± 20.6	59.2 ± 18.1	<0.001*
Total cholesterol, mg/dl	195.4 ± 44.3	186.3 ± 31.8	0.419
Triglycerides, mg/dl	164.6 ± 101.6	110.1 ± 53.9	<0.001*
HOMA-IR	5.8 ± 10.5	2.6 ± 2.5	0.021*
CRP, mg/l	3.0 ± 5.5	2.6 ± 3.9	0.010*
Lactic acid, μmol/g	0.40 ± 1.62	0.08 ± 0.13	0.970
Propionic acid, μmol/g	9.8 ± 8.4	9.6 ± 11.5	0.079
Butyric and isobutyric acid, μmol/g	9.1 ± 8.4	9.5 ± 11.3	0.373
Acetic acid, μmol/g	26.5 ± 27.4	26.0 ± 30.1	0.204
Valeric acid, μmol/g	1.5 ± 5.0	1.0 ± 1.1	0.547
Isovaleric acid, μmol/g	0.56 ± 0.65	0.29 ± 0.56	0.014*

Significant differences after the Benjamini-Hochberg correction.
Peripheral blood markers were not available for 2 individuals with schizophrenia.
Abbreviations: CRP, C-reactive protein; HDL, high-density lipoproteins; HOMA-IR, the Homeostasis Model Assessment of Insulin Resistance; LDL, low-density lipoproteins.

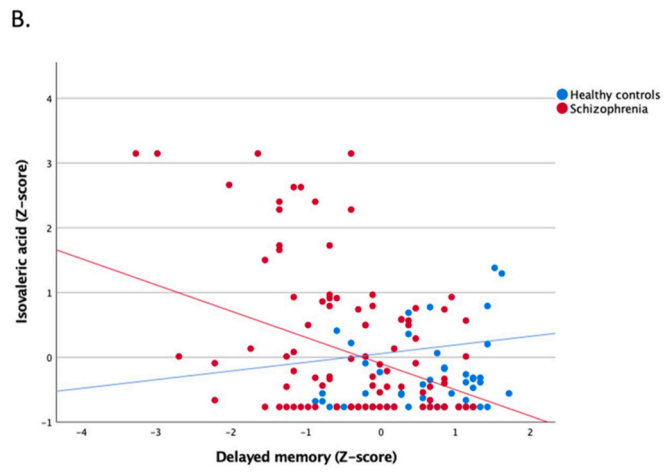


Fig. 1. Fecal levels of isovaleric acid in patients with schizophrenia and healthy controls (A), and their correlations with delayed memory (B). Mean levels and standard deviations are shown (A). Abbreviations: HCs, healthy controls; SCZ, schizophrenia.

were associated with significantly lower RBANS scores of immediate and delayed memory in patients with schizophrenia. In healthy controls, higher levels of acetic acid were associated with lower RBANS scores of the language domain. These correlations were further tested using linear regression analysis. The levels of CRP and HDL were added to the analyses modeling the effects of isovaleric acid levels on delayed memory as the levels of these biochemical parameters differed significantly between both groups of participants and were significantly correlated with the RBANS scores of immediate and delayed memory (Supplementary Table 1). In case of immediate memory, the levels of CRP were included in multiple linear regression analysis as only this variable was associated with the RBANS scores of immediate memory. In turn, the RBANS language scores were not significantly correlated with any biochemical parameters, BMI, or the aMED score. Therefore, covariates were not added to this model. Two participants were identified to show outlier values of cognitive performance and the levels of isovaleric acid (among individuals with schizophrenia).

Multiple linear regression analysis revealed a significant negative correlation of isovaleric acid levels with the RBANS scores of delayed memory (Table 5). However, this correlation appeared to be not significant when the interaction terms were added to the model. In turn, the association of the interaction between group status and isovaleric acid levels with the RBANS scores of delayed memory was significant in the models testing for the effects of two-way and three-way interactions.

Table 3
Correlations of the levels of lactic acid and SCFAs with the levels of metabolic parameters, cigarette smoking and dietary habits.

Group	Variable	Lactic acid	Propionic acid	Butyric and isobutyric acid	Acetic acid	Valeric acid	Isovaleric acid
SCZ	BMI	r = −0.052	r = 0.012	r = 0.043	r = −0.115	r = 0.036	r = 0.071
	Total cholesterol	r = −0.066	r = −0.136	r = −0.133	r = −0.255	r = 0.062	r = 0.106
	Triglycerides	r = 0.079	r = −0.092	r = −0.154	r = −0.094	r = 0.001	r = 0.232
	LDL	r = −0.037	r = −0.248	r = −0.234	r = −0.351*	r = −0.058	r = 0.050
	HDL	r = −0.011	r = 0.062	r = 0.023	r = 0.057	r = 0.069	r = −0.185
	HOMA–IR	r = −0.078	r = −0.098	r = −0.095	r = −0.021	r = −0.115	r = −0.040
	CRP	r = −0.071	r = 0.173	r = 0.326	r = 0.254	r = 0.368*	r = 0.439*
	FTND	r = −0.022	r = 0.022	r = −0.035	r = 0.033	r = −0.036	r = 0.073
	aMED	r = −0.001	r = −0.071	r = 0.043	r = −0.115	r = 0.007	r = −0.063
HCs	BMI	r = 0.338	r = 0.166	r = 0.227	r = 0.163	r = 0.096	r = −0.090
	Total cholesterol	r = 0.003	r = 0.050	r = 0.072	r = 0.047	r = 0.174	r = 0.136
	Triglycerides	r = 0.070	r = 0.507*	r = 0.455*	r = 0.441*	r = 0.408*	r = 0.037
	LDL	r = 0.132	r = 0.017	r = 0.035	r = 0.003	r = 0.150	r = 0.216
	HDL	r = −0.199	r = −0.266	r = −0.243	r = 0.191	r = −0.211	r = −0.022
	HOMA–IR	r = 0.269	r = 0.073	r = 0.058	r = −0.017	r = 0.100	r = 0.144
	CRP	r = 0.178	r = 0.398*	r = 0.394*	r = 0.402*	r = 0.275	r = −0.004
	FTND	r = 0.130	r = −0.134	r = −0.142	r = −0.138	r = −0.133	r = 0.189r
	aMED	r = −0.068	r = 0.076	r = 0.193	r = 0.153	r = 0.198	r = −0.081

*Significant correlations after the Benjamini–Hochberg correction.
Abbreviations: aMED, adherence to Mediterranean diet; BMI, body mass index; CRP, C–reactive protein; HCs, healthy controls; HDL, high–density lipoproteins; LDL, low–density lipoproteins; SCZ, schizophrenia.

Table 4
Correlations of the levels of lactic acid and SCFAs with clinical characteristics and cognitive performance.

Group		Lactic acid	Propionic acid	Butyric and isobutyric acid	Acetic acid	Valeric acid	Isovaleric acid
SCZ	PANSS–N	r = −0.070	r = −0.062	r = −0.155	r = −0.026	r = 0.010	r = 0.046
	CDSS	r = −0.043	r = 0.041	r = 0.066	r = 0.061	r = −0.126	r = −0.092
	SOFAS	r = 0.124	r = −0.028	r = 0.069	r = −0.075	r = −0.102	r = −0.157
	CPZeq	r = −0.021	r = −0.062	r = −0.075	r = −0.125	r = −0.012	r = 0.131
	Immediate memory	r = 0.148	r = 0.060	r = 0.064	r = 0.075	r = −0.212	r = −0.278*
	Visuospatial/constructional abilities	r = −0.040	r = 0.181	r = 0.130	r = 0.191	r = 0.097	r = −0.035
	Language	r = −0.048	r = 0.024	r = −0.012	r = 0.032	r = 0.112	r = −0.057
	Attention	r = 0.069	r = 0.009	r = −0.030	r = 0.001	r = −0.146	r = −0.117
	Delayed memory	r = −0.051	r = 0.015	r = −0.001	r = 0.062	r = −0.199	r = −0.265*
HCs	Immediate memory	r = −0.198	r = −0.010	r = −0.071	r = −0.087	r = −0.051	r = 0.017
	Visuospatial/constructional abilities	r = 0.030	r = −0.103	r = −0.168	r = −0.206	r = −0.037	r = 0.117
	Language	r = −0.143	r = −0.184	r = −0.263	r = −0.274*	r = −0.192	r = −0.078
	Attention	r = −0.130	r = −0.143	r = −0.172	r = −0.223	r = −0.161	r = 0.008
	Delayed memory	r = −0.113	r = 0.149	r = 0.062	r = 0.075	r = 0.057	r = 0.058

*Significant correlations after the Benjamini–Hochberg correction.
Abbreviations: HCs, healthy controls; SCZ, schizophrenia.

Table 5
Results of multiple linear regression analysis modeling the effects of isovaleric acid levels on cognitive performance.

Dependent variable	Model	Independent variable	Beta	p	R^2
Language	Main effects	Group	−0.438	<0.001	0.187*
		Acetic acid	−0.073	0.320	
	Main effects and two-way interactions	Group	−0.437	<0.001	0.199
		Acetic acid	−0.214	0.053	
		Group × acetic acid	0.208	0.071	
Immediate memory	Main effects	Group	−0.453	<0.001	0.355*
		Isovaleric acid	−0.168	0.059	
		CRP	−0.211	0.002	
	Main effects and two-way interactions	Group	−0.463	<0.001	0.358
		Isovaleric acid	−0.140	0.505	
		CRP	−0.124	0.420	
		Group × isovaleric acid	−0.135	0.893	
		Group × CRP	−0.096	0.532	
	Main effects, two-way interactions, and three-way interactions	Group	−0.473	<0.001	0.362
		Isovaleric acid	−0.140	0.505	
		CRP	−0.124	0.420	
		Group × isovaleric acid	−0.093	0.663	
		Group × CRP	−0.018	0.919	
		Group × isovaleric acid × CRP	0.115	0.347	
Delayed memory	Main effects	Group	−0.383	<0.001	0.333*
		Isovaleric acid	−0.310	<0.001	
		CRP	−0.004	0.959	
		HDL	0.107	0.124	
	Main effects and two-way interactions	Group	−0.475	<0.001	0.379*
		Isovaleric acid	0.239	0.249	
		CRP	0.056	0.732	
		HDL	0.121	0.409	
		Group × isovaleric acid	−0.463	0.001	
		Group × CRP	−0.059	0.714	
		Group × HDL	−0.005	0.969	
	Main effects, two-way interactions, and three-way interactions	Group	−0.465	<0.001	0.383
		Isovaleric acid	0.239	0.103	
		CRP	0.056	0.733	
		HDL	0.121	0.411	
		Group × isovaleric acid	−0.501	0.020	
		Group × CRP	−0.135	0.459	
		Group × HDL	−0.003	0.983	
		Group × isovaleric acid × CRP	−0.113	0.367	
		Group × isovaleric acid × HDL	−0.007	0.925	

*significant F change ($p < 0.05$).
Abbreviations: CRP, C-reactive protein; HDL, high-density lipoproteins.

There were no significant associations of isovaleric acid levels with the RBANS scores of immediate memory in multiple regression analyses. Similarly, the correlation of acetic acid levels with the RBANS language scores was not significant. In neither of models, significant associations between the RBANS scores and interactions with the levels of HDL or CRP were found.

4. Discussion

Findings from the present study indicate that the fecal levels of isovaleric acid are elevated in patients with schizophrenia during remission of positive symptoms. In agreement with these observations, Liang et al. (2022a) revealed elevated fecal and serum levels of branched-chain SCFAs (isobutyric acid and isovaleric acid) in patients with schizophrenia. It is important to note that branched SCFAs are generated exclusively from transamination of branched-chain amino acids. In turn, straight-chain SCFAs, which were not elevated, may be derived from carbohydrates (such as dietary fibers). Consequently, Liang et al. (2022a) hypothesized that these findings might indicate enhanced gut catabolism of amino acids in schizophrenia. Despite the fact that antipsychotics have been reported to impact gut microbiome (Seeman, 2021), Liang et al. (2022a) did not report correlations between the dose of antipsychotics and microbial metabolism of macronutrients. This observation is also in agreement with our findings that the CPZeq is not significantly associated with the levels of SCFAs. Similar observations were made by Li et al. (2022), who found elevated levels of total SCFAs, acetic acid and the acetic acid/propionic acid ratio in drug-naïve patients with first-episode schizophrenia.

Our study also demonstrated significant association of higher levels of isovaleric acid with worse performance of delayed memory in patients with schizophrenia, but not in healthy controls. A lack of significant correlation with cognition in healthy controls might be due to the fact that individuals with schizophrenia showed significantly higher levels of isovaleric acid. This might suggest that the association of isovaleric acid levels with cognitive performance appears at higher concentrations. Also, it is important to note that individuals with schizophrenia presented worse cognitive performance. Importantly, the associations of the levels of isovaleric acids with impairments of delayed memory cannot be explained by their interactions with CRP levels or other metabolic parameters. Findings from the study by Li et al. (2022) indicate that elevated levels of SCFAs in drug-naïve patients with schizophrenia contribute to cognitive impairment. Indeed, the authors found that a higher acetic acid/propionic acid ratio is associated with impairments of working memory, reasoning and problem-solving. However, the exact mechanisms underlying our observations with respect to cognitive performance remain unclear. One scenario is that isovaleric acid can impact cognition through its interactions with adenosine A1 receptors (Lacher et al., 2007). Another potential explanation is that isovaleric acid can contribute to the activation of the hypothalamic-pituitary-adrenal (HPA) axis. Indeed, it has been reported that depressed patients show increased levels of isovaleric acid that are associated with higher cortisol levels (Szczesniak et al., 2016). Several lines of evidence indicate dysfunction of the HPA axis in patients with psychosis that might already be present during preclinical stages, and manifests in pituitary enlargement, higher blood cortisol levels, blunted cortisol awakening response as well as blunted cortisol response to stress (Misiak et al., 2020). Prolonged exposure to elevated cortisol levels may give rise to morphological and molecular alterations, decreased neurogenesis and disrupted synaptic plasticity in the hippocampus that is responsible for learning and memory processes (Kim et al., 2015). Also, it has been reported that valeric acid contributes to neuroinflammation as well as impairments of learning and memory processes in mice after surgery through the interactions with complement signaling (Lai et al., 2021).

The present study has some limitations which need to be highlighted during the interpretation of our findings. First, we did not perform the analysis of SCFAs levels in the peripheral blood. Second, the use of a convenience sample may not provide generalizability of findings over the whole population of patients with schizophrenia. Third, due to a cross-sectional design, it is not possible to indicate causal associations. Moreover, we used a subjective method (self-reports) to assess dietary habits, and the recall bias cannot be excluded due to assessment of food intake over the preceding 12 months. Another limitation is that we did not record lifestyle variables, e.g., physical activity, sedentary behaviors, cannabis use and alcohol consumption, and thus latent confounding by these factors cannot be excluded (Caslin et al., 2021; Tzemah Shahar et al., 2020). Finally, the analysis of subclinical inflammation was limited to assessment of CRP levels.

Our results indicate that individuals with schizophrenia show elevated levels of isovaleric acid that are associated with subclinical inflammation as well as impairments of delayed memory. However, it remains unknown as to whether these observations are associated with

illness progression. In this regard, studies of individuals with first–episode psychosis are needed to inform as to whether observed associations represent trait or state characteristics of schizophrenia. Future studies with comprehensive assessment of gut microbiota composition and function as well as the analysis of SCFAs in the peripheral blood and fecal samples might also provide further insights. Furthermore, longitudinal and experimental studies are needed to explain the mechanisms linking impaired metabolism of SCFAs with schizophrenia.

Author contributions

Krzysztof Kowalski: Conceptualization, Methodology, Writing - Original Draft, Formal analysis, Resources, Data Curation, Project administration, Funding acquisition Bogumiła Szponar: Investigation, Patrycja Bochen: Investigation, Paulina Żebrowska: Investigation, Łukasz Łaczmański: Investigation, Jerzy Samochowiec: Writing - Review & Editing, Błażej Misiak: Writing – Conceptualization, Methodology, Writing - Review & Editing, Software, Formal analysis, Visualization, Supervision.

Funding source

This study received funding from the research grant awarded by Wroclaw Medical University, Wroclaw, Poland (grant number: SUBK. C230.22.050).

Declaration of competing interest

None to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpsychires.2023.07.042.

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Update

Journal of Psychiatric Research
Volume 169, Issue , January 2024, Page 191

DOI: https://doi.org/10.1016/j.jpsychires.2023.11.001

Journal of Psychiatric Research 169 (2024) 191–191

Contents lists available at ScienceDirect

Journal of Psychiatric Research

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Corrigendum to "Altered levels of fecal short-chain fatty acids are associated with subclinical inflammation and worse cognitive performance in patients with schizophrenia" [J. Psychiatr. Res. 165 (2023) 298–304]

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The authors regret that the original version of the article included incomplete Fig. 1. The correct version of Fig. 1 is published in this corrigendum.

The authors would like to apologise for any inconvenience caused.

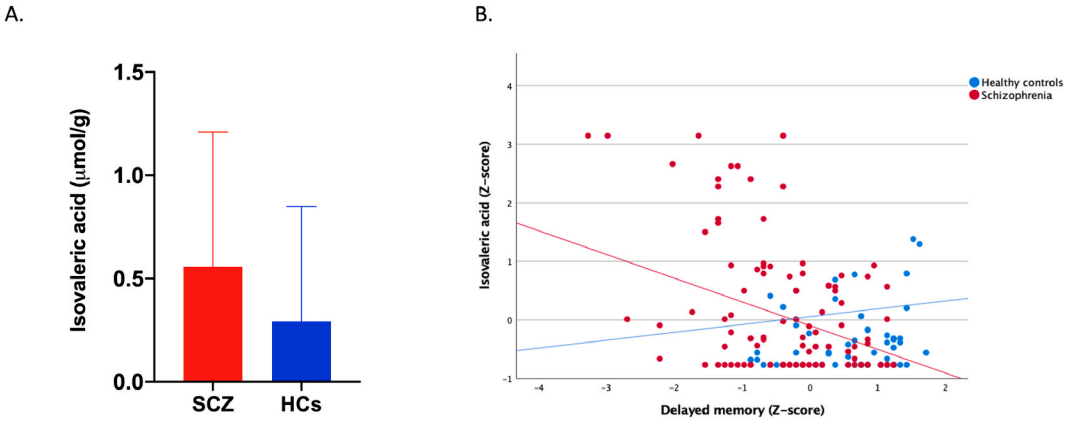


Fig. 1. Fecal levels of isovaleric acid in patients with schizophrenia and healthy controls (A), and their correlations with delayed memory (B). Mean levels and standard deviations are shown (A). Abbreviations: HCs, healthy controls; SCZ, schizophrenia.

DOI of original article: https://doi.org/10.1016/j.jpsychires.2023.07.042.
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https://doi.org/10.1016/j.jpsychires.2023.11.001

Available online 1 December 2023
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